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Anopheles funestus resistant to pyrethroid insecticides in South Africa

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Abstract. Northern Kwazulu/Natal (KZN) Province of South Africa borders on southern Mozambique, between Swaziland and the Indian Ocean. To control malaria vectors in KZN, houses were sprayed annually with residual DDT 2 g/m^2 until 1996 when the treatment changed to deltamethrin 20–25 mg/m^2. At Ndumu (27°02′ S, 32°19′ E) the recorded malaria incidence increased more than six-fold between 1995 and 1999. Entomological surveys during late 1999 found mosquitoes of the *Anopheles funestus* group (Diptera: Culicidae) resting in sprayed houses in some sectors of Ndumu area. This very endophilic vector of malaria had been eliminated from South Africa by DDT spraying in the 1950s, leaving the less endophilic *An. arabiensis* Patton as the only vector of known importance in KZN. Deltamethrin-sprayed houses at Ndumu were checked for insecticide efficacy by bioassay using susceptible *An. arabiensis* (laboratory-reared) that demonstrated 100% mortality. Members of the *An. funestus* group from Ndumu houses (29 males, 116 females) were identified by the rDNA PCR method and four species were found: 74 *An. funestus* Giles *sensu stricto,* 34 *An. parensis* Gillies, seven *An. rivulorum* Leeson and one *An. leesoni* Evans. Among *An. funestus s.s.* females, 5.4% (4/74) were positive for *Plasmodium falciparum* by ELISA and PCR tests. To test for pyrethroid resistance, mosquito adults were exposed to permethrin discriminating dosage and mortality scored 24 h post-exposure: survival rates of wild-caught healthy males were 5/10 *An. funestus*, 1/9 *An. rivulorum* and 0/2 *An. parensis*; survival rates of laboratory-reared adult progeny from 19 *An. funestus* females averaged 14% (after 1 h exposure to 1% permethrin 25 : 75 cis : trans on papers in WHO test kits) and 27% (after 30 min in a bottle with 25 µg permethrin 40 : 60 cis : trans). *Anopheles funestus* families showing >20% survival in these two resistance test procedures numbered 5/19 and 12/19, respectively. Progeny from 15 of the families were tested on 4% DDT impregnated papers and gave 100% mortality. Finding these proportions of pyrethroid-resistant *An. funestus*, associated with a malaria upsurge at Ndumu, has serious implications for malaria vector control operations in southern Africa.

Key words. *Anopheles funestus*, deltamethrin, malaria vector, molecular identification, permethrin, pyrethroid resistance, susceptibility tests, South Africa.

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Introduction

Chemical insecticides have been used against *Anopheles* mosquitoes in malaria control programmes for over 60 years, with varying success (Park Ross, 1936; Harrison, 1978). Pyrethrum extracted from flowers was sprayed in houses as a short-term knockdown insecticide (De Meillon, 1936) and was superseded by the more residual organochlorines (DDT, dieldrin, γHCH) after the Second World War. Widespread effective use of organochlorine insecticides for agricultural purposes as well as antimalaria house-spraying (Pampana, 1969) resulted in the emergence of resistant strains of malaria vector mosquitoes (Brown & Pal, 1973) and, together with environmental concerns, this led to the withdrawal of some insecticides (Brown, 1986). Alternative insecticide classes (carbamates and organophosphates) were introduced for use against vectors and pests of public health importance (Fontaine, 1983). Most recently the development of photo-stable synthetic pyrethroids led to several of these products, such as deltamethrin (Chavasse & Yap, 1997), replacing DDT for house-spraying in some malaria control programmes during the 1990s. More importantly, pyrethroids are currently the only practical insecticides for impregnating bednets, a widely promoted vector control intervention (Curtis, 1990; Lengeler *et al*., 1996).

Pyrethroid resistance or tolerance in the malaria vector *Anopheles gambiae* Giles *sensu stricto* has been reported from both west and east Africa (Elissa *et al*., 1993; Vulule *et al*., 1994, 1996). Recent sampling in Burkina Faso and the Ivory Coast (Martinez-Torres *et al*., 1998; Chandre *et al*., 1999a, b, c) and Kenya (Vulule *et al*., 1999; Ranson *et al*., 2000) revealed extensive pyrethroid resistance in wild populations of *An. gambiae*, involving at least three mechanisms (increased esterase and hydrolase metabolism, and knockdown resistance, *kdr*).

Anopheles funestus Giles *sensu stricto* is a major vector of malaria, widespread in tropical Africa; being highly anthropophilic and endophilic, this species is amenable to control by house-spraying with residual insecticides (Gillies & De Meillon, 1968; Kouznetsov, 1977). *Anopheles funestus* was eliminated from South Africa by DDT house-spraying in the 1950s (Sharp & Le Sueur, 1996), since when it has only once been reported again in this country (De Meillon *et al*., 1977). However, *An. funestus* remains abundant nearby in southern Mozambique (Charlwood *et al*., 1998), where it transmits malaria perennially (Mendis *et al*., 2000). There have been only a few old reports of insecticide resistance in *An. funestus:* to dieldrin in Benin, Cameroon, Ghana, Kenya, Mali and Nigeria (Brown, 1986), as well as to DDT, malathion and fenitrothion in Mali (Touré, 1982). There is no previous report of pyrethroid resistance in *An. funestus* and, despite long-term house-spraying, no signs of insecticide resistance have been detected previously in *An. funestus* from southern Africa.

Anopheles funestus belongs to a group of at least nine morphologically similar species (Gillies & De Meillon, 1968; Gillies & Coetzee, 1987). Four of these (*An. brucei* Service, *An. confusus* Evans & Leeson, *An. leesoni* Evans and *An. rivulorum* Leeson) can be identified from characters of the egg and larval morphology, one is known only in the adult stage (*An. fuscivenosus* Leeson) and four species comprising the *An. funestus* subgroup (*An. aruni* Sobti, *An. funestus s.s., An. parensis* Gillies and *An. vaneedeni* Gillies & Coetzee) are morphologically almost indistinguishable in all life stages (Gillies & Coetzee, 1987). Cytogenetic studies showed that banding arrangements of the polytene chromosomes could be used to separate these species (Green & Hunt, 1980; Green, 1982) but, because of the time and skill needed to interpret chromosomal banding patterns, this method has never been used for applied field studies of the *An. funestus* group. Recently we developed molecular methods (see below and Koekemoer *et al*., 1999) to distinguish between all five members of the group known from South Africa, i.e. *An. funestus s.s., An. leesoni, An. parensis, An. rivulorum* and *An. vaneedeni*.

In northern Kwazulu/Natal (KZN) Province of South Africa, bordering Mozambique between Swaziland and the Indian Ocean (Fig. 1), malaria has increased over the past four years despite continued and efficient house spraying for malaria control. From 1995 to 1999 the numbers of malaria cases recorded annually for Kwazulu/Natal were 4117, 10535, 11425, 14575 and 27238, respectively (KZN Department of Health statistics). This paper presents evidence that *An. funestus* has reappeared in the Ndumu area of northern KZN, where this species now occurs inside sprayed houses, shows partial resistance to pyrethroid insecticides and is responsible for malaria transmission. Moreover, we determined that *An. funestus s.s.* is not the only member of the *An. funestus* group to be found resting inside houses at Ndumu.

Materials and Methods

Mosquito collections

The study area of Ndumu (27°02′ S, 32°19′ E) in northern KZN borders on southern Mozambique (Fig. 1). Communities around the Ndumu Game Reserve are covered by vector control operations of the KZN malaria control programme, whereby all houses are sprayed annually with the pyrethroid insecticide deltamethrin (5% wettable powder applied at the target rate of 20–25 mg ai/m^2). Deltamethrin was introduced in 1996 after ~40 years of house-spraying with DDT. Within the Game Reserve, malaria control is the responsibility of the Nature Conservation authorities, but not all structures are sprayed regularly.

Following WHO (1975) procedures, window exit traps (WT) and pyrethrum knockdown spray catches (PSC) were used to monitor adult *Anopheles* mosquitoes from deltamethrin-sprayed houses of the malaria control programme area and unsprayed houses within Ndumu Game Reserve. Between 11 October and 12 November 1999, in 28

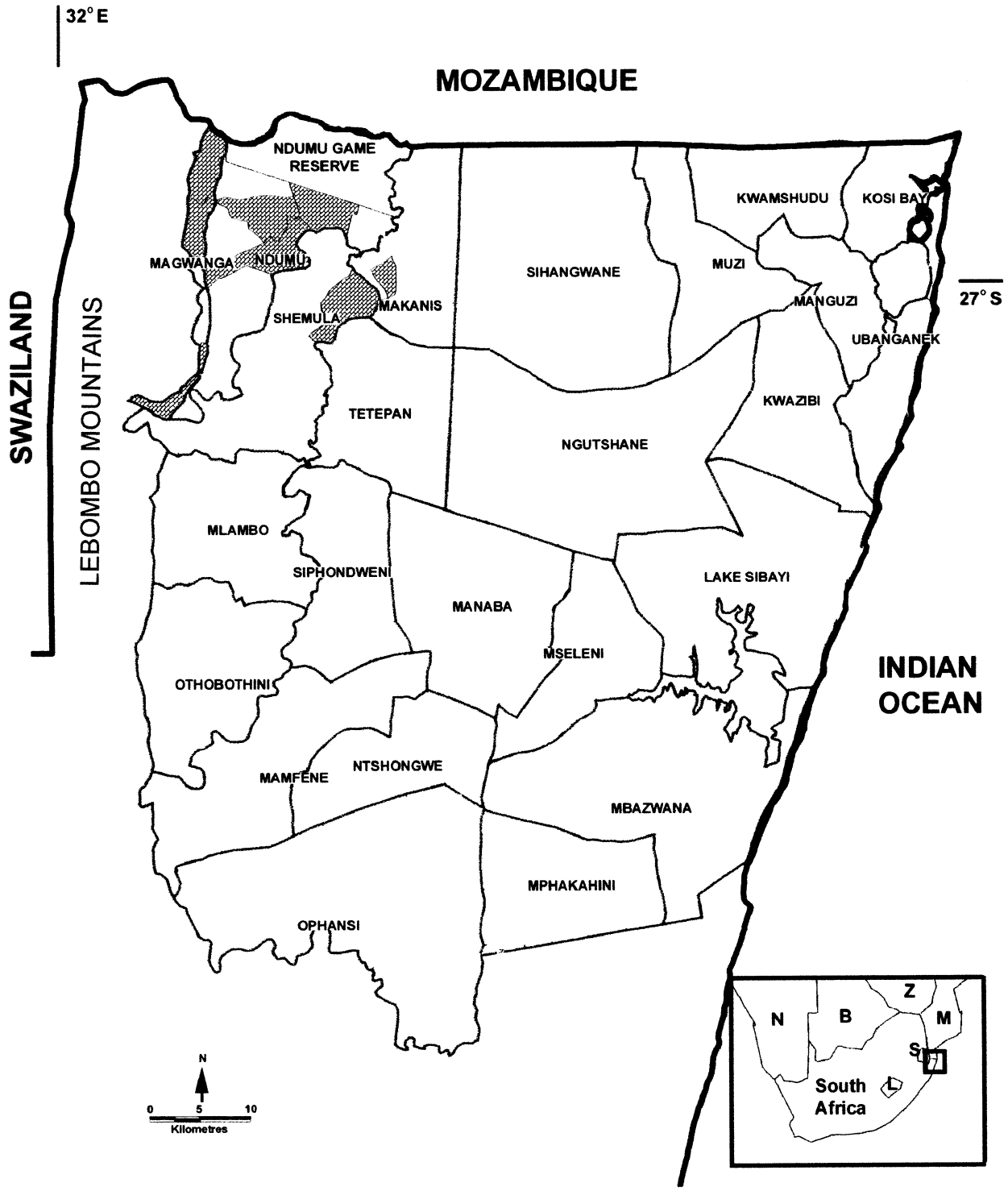


Fig. 1. Map of northern Kwazulu/Natal bordering on Mozambique, showing sector boundaries. Shaded areas are where collections were carried out within the sectors. Insert map code: B – Botswana, L – Lesotho, M – Mozambique, N – Namibia, S – Swaziland, Z – Zimbabwe.

homesteads 30 WTs were operated for 25 nights; thereafter in 46 homesteads 57 WTs were operated for 15 nights from 15 November to 3 December 1999.

PSCs were implemented in eight houses, where WT collections yielded mosquitoes of the *An. funestus* group. Dead mosquitoes, mostly collected by PSC, were stored

Table 1. Collections, identifications and resistance data of *Anopheles funestus* group mosquitoes from northern Kwazulu/Natal, South Africa

Locality	No. houses	Collection Method	Species	No. female	No. male	*P. falciparum* positive	Resistance status [1]
Makanis	2	Window traps	*An. funestus*	6	–	0	not tested
Shemula	1	Knockdown	*An. rivulorum*	2	–	0	not tested
Magwangwa	3	Knockdown	*An. funestus*	1	–	0	not tested
		Window traps	*An. funestus*	1	–	0	not tested
Ndumu Reserve	4	Window traps	*An. rivulorum*	1	–	0	not tested
		Indoor resting	*An. rivulorum*	4	9	0	1 resistant, 8 susceptible, males
		Indoor resting	*An. funestus*	1	–	0	not tested
Ndumu	40	Window traps	*An. parensis*	25	6	0	2 susceptible males
		Knockdowns	*An. parensis*	9	–	0	not tested
		Window traps	*An. funestus*	60	14	3	5 resistant, 2 moribund, 3 susceptible males
		Knockdowns	*An. funestus*	5	–	1	not tested
		Window trap	*An. leesoni*	1	–	0	not tested

[1] Susceptible *An. arabiensis* colony (KGB strain) used as controls.

desiccated with silica gel for identification and malaria parasite infection tests.

Bioassays of deltamethrin residual efficacy in sprayed houses

Laboratory-reared *An. arabiensis* adults (KGB strain, originated from the Zambezi Valley, Zimbabwe) known to be susceptible to deltamethrin, i.e. 100% mortality on 0.05% deltamethrin (WHO, 1998), were used to bioassay the insecticide deposits inside eight houses (deltamethrin-sprayed ~6 weeks previously) where anophelines had been detected by PSC or WT. Three plastic bioassay cones (WHO, 1975) were attached to the inner surface of walls of the house, in randomly chosen positions at various heights. For each bioassay, 20 3-day-old female mosquitoes were introduced into the cone. After 30 min exposure, mosquitoes were removed from the cone and placed in polystyrene cups for a 24 h holding period. Control mosquitoes were monitored for the same period. Mortality was recorded at the end of 30 min exposure and survival noted after 24 h.

Pyrethroid susceptibility of mosquitoes collected live from window traps

Male mosquitoes, held for 24 h after collection from WTs, were tested by the CDC bottle bioassay method of Brogdon & McAllister (1998). This involved gently aspirating the mosquitoes into Wheaton bottles (250 ml capacity) pre-treated with permethrin (40 : 60 cis : trans isomer ratio, permethrin 25 µg/bottle, applied as 0.25% aqueous 10 µl/bottle, with 1 ml acetone as carrier). Treated bottles were transported to the field in a dark container, to avoid any possible photodegradation of permethrin. After 30 min exposure in a permethrin-treated bottle, mosquitoes were transferred to a recovery cage, with access to 10% sugar solution for nourishment, and mortality was recorded 24 h post-exposure. We previously ascertained that this permethrin treatment kills susceptible *An. arabiensis* with 30 min exposure.

Progeny were reared from individual egg batches (families) laid by females of the *An. funestus* group collected from WTs. Adult sugar-fed F_1 males and females (aged 1-3 days, non-bloodfed) were assessed for susceptibility/resistance to permethrin by two test procedures: the bottle bioassay, as above, and the standard WHO (1975) test kit procedure. Mosquitoes were exposed by tarsal contact for 1 h to filter papers impregnated with 1% permethrin 25 : 75 cis : trans in silicone oil (Dow Corning 556); this dose had been predetermined as diagnostic for resistance in *An. gambiae* (Duchon *et al*., 1997). Susceptible *An. arabiensis* (KGB strain) or *An. quadriannulatus* (Theobald) (SKUQUA strain, originating from the Kruger National Park, South Africa) were used as controls in all assays.

Identification of mosquitoes and malaria parasites

Individual mosquitoes were first identified by morphological keys (Gillies & De Meillon, 1968; Gillies & Coetzee, 1987). Those belonging to the *An. funestus* group were subjected to PCR-based single strand conformation polymorphism (SSCP) using one leg (Koekemoer *et al*., 1999) for distinguishing *An. funestus*, *An. leesoni*, *An. rivulorum* and *An. parensis/vaneedeni*. Because *An. parensis* has the same SSCP configuration as *An. vaneedeni*, these two species were separated by subsequent PCR of a segment of rDNA (unpublished data).

All wild females were individually tested for the presence of *Plasmodium falciparum* malaria parasites using the ELISA method of Wirtz *et al*. (1987) and/or the PCR method of Snounou *et al*. (1993).

Results

Bioassays inside houses

Insecticide efficacy was confirmed by bioassays on treated walls of eight deltamethrin-sprayed houses. From 24 cone tests (3 × 20 mosquitoes per house), all of ~480 susceptible *An. arabiensis* were knocked down within 30 min exposure and died < 24 h post-exposure to treated walls in deltamethrin-sprayed houses, whereas untreated controls subjected to similar handling showed < 1% mortality.

Mosquito identifications and infectivity

Collections from 50 houses (by PSC and WT) in five localities at Ndumu yielded 29 males and 116 females of the *An. funestus* group. DNA assays identified four species: *An. funestus*, *An. leesoni*, *An. parensis*, *An. rivulorum* (Table 1). No specimens of *An. arabiensis* were collected. Among 130 females tested for *P. falciparum* circumsporozoite protein and/or parasite DNA, four specimens (3.1%) were positive. These four females were all identified as *An. funestus s.s.*, giving an infection rate of 5.4% for this species (Table 1).

Susceptibility tests

According to criteria adopted by the WHO (1998), < 80% mortality 24 h post-exposure indicates resistance. By this criterion, 5/19 families of *An. funestus s.s.* (i.e. 26%) showed evidence of resistance after 1 h exposure to 1% permethrin 25 : 75 papers (Table 2). Bottle bioassays, with a less well standardized diagnostic dose, indicated resistance in 63% (12/19) of these families. Only three families (coded NDF 88, 101 and 200 in Table 2) appeared to be resistant by both test methods. The overall survival (i.e. resistance) rates following exposure to a putative diagnostic dose of permethrin was 26.8% (120/164) for the bottle bioassay and 13.7% (210/244) for the impregnated papers. When the result indicated resistance, however, the knock-down effect was marked and recovery occurred during the 24 h holding period, suggesting a metabolic mechanism of resistance.

A total of 21 wild-caught male anophelines were bioassayed in permethrin-treated bottles. Species identification and exposure results are given in Table 1: mortality rates within 24 h were 2/2 *An. parensis* and 8/9 *An. rivulorum* but only 5/10 *An. funestus s.s.*, whereas susceptible *An. arabiensis* exposed to the same treatment all died and those in bottles treated only with acetone all survived (negative and positive controls, respectively). Results with males therefore indicate permethrin resistance in 50% of *An. funestus s.s.*, whereas *An. parensis* and *An. rivulorum* were apparently susceptible.

Fifteen families of *An. funestus s.s.* were tested on 4% DDT impregnated papers and were 100% susceptible (Table 2).

Discussion

The recent upsurge of malaria in northern KZN may be attributed to several factors, including good summer rainfall and mild winters during 1996–99, influx of gametocyte carriers to KZN from Mozambique and rising *P. falciparum* resistance to chloroquine (Freese *et al*., 1991) and sulphadoxine pyrimethamine (unpublished report, 2000, preliminary data presented to Malaria Advisory Group of SA Ministry of Health). The doubling of malaria cases from 1998 to 1999, however, is undoubtedly a consequence of reinvasion by the highly anthropophilic *An. funestus* after its absence for several decades. This efficient vector caused major outbreaks of malaria after reinvading the plateau of Madagascar during the 1980s (Lepers *et al*., 1990; Laventure *et al*., 1995).

There are two possible reasons for the absence of *An. arabiensis* from the collections. Firstly, this species is highly seasonal and only appears in detectable numbers after the summer rains get underway, i.e. from January to May (Mendis *et al*., 2000). Secondly, no pyrethroid resistance has been detected in local populations of *An. arabiensis* with the result that those individuals entering sprayed houses would, in all likelihood, not survive to be collected by indoor trapping techniques.

Pyrethroid susceptibility tests were made on mosquitoes of the *An. funestus* group from houses recently sprayed with residual deltamethrin and the treatment was proved to be effective against susceptible *An. arabiensis* in bioassays. For convenience, initial tests employed the bottle bioassay of Brogdon & McAllister (1998), using a tentative diagnostic dosage of permethrin for detection of resistance. Confirmation of pyrethroid resistance depended on the WHO test kit procedure for *An. gambiae*, but using 1% permethrin 25 : 75 papers supplied by a WHO Collaborating Centre (LIN/IRD), instead of the standard 0.75% permethrin 40 : 60 (WHO, 1998). Paired samples from the same families of *An. funestus* were exposed to each assay in order to provide comparable data. Some families of *An. funestus* were clearly resistant to permethrin by both test procedures. Other discrepancies in mortality results may be due to the different dose (time × concentration) applied in each test procedure, different isomer ratios of the permethrin used, silicone vs. acetone/water as carriers of the permethrin, uneven coating of the bottles, or random selection of different genotype frequencies in the subfamilies tested.

One drawback to working with *An. funestus* was the lack of baseline data on fully susceptible strains for comparison with field populations. No laboratory colony of *An. funestus* exists on which such an evaluation could be carried out, and no susceptibility tests have been made previously on F_1 progeny of wild females of *An. funestus*. In this study we employed susceptible *An. arabiensis* and *An. quadriannulatus* (two members of the *An. gambiae* complex) as our reference strains, but there are no supporting data to confirm that the discriminating dose for the *An. gambiae* complex is also applicable to *An. funestus*. Although survival on 1% permethrin is strong evidence of pyrethroid resistance, further studies need to be conducted to establish discriminating dosages for

Table 2. Mortality data for F_1 adults (males and females) of 19 families of *Anopheles funestus s.s.* (identified by SSCP) from parent females collected inside houses in northern Kwazulu/Natal, South Africa. See text for details of test procedures; mortality recorded 24 h post-exposure.

Code number	CDC bottle bioassay % (no. dead/tested) Permethrin	WHO impregnated papers % (no. dead/tested) Permethrin	DDT
R2	88 (7/8)	100 (5/5)	100 (4/4)
NDF 88	50 (3/6)	69 (11/16)	100 (14/14)
NDF 89	100 (6/6)	70 (14/20)	100 (14/14)
NDF 90	83 (5/6)	87 (13/15)	–
NDF 93	100 (5/5)	40 (2/5)	–
NDF 101	60 (3/5)	73 (11/15)	100 (11/11)
NDF 133	75 (6/8)	100 (10/10)	100 (5/5)
NDF 139	56 (9/16)	83 (5/6)	100 (14/14)
NDF 140	86 (6/7)	90 (9/10)	100 (5/5)
NDF 147	63 (5/8)	100 (9/9)	–
NDF 152	50 (2/4)	83 (5/6)	–
NDF 155	75 (9/12)	85 (10/13)	100 (5/5)
NDF 159	75 (12/16)	100 (13/13)	100 (11/11)
NDF 169	25 (1/4)	93 (14/15)	100 (8/8)
NDF 170	100 (6/6)	100 (14/14)	100 (18/18)
NDF 179	78 (14/18)	100 (13/13)	100 (10/10)
NDF 195	89 (8/9)	100 (25/25)	100 (17/17)
NDF 197	71 (10/14)	100 (22/22)	100 (17/17)
NDF 200	67 (4/6)	67 (8/12)	100 (11/11)
Pooled	73 (120/164)	86 (210/244)	100 (164/164)
Positive control[1]	5 (7/166)	4 (8/202)	4 (7/166)
Negative control[2]	100 (189/189)	99 (168/169)	100 (147/147)

[1] Positive control: susceptible *An. quadriannulatus* (SKUQUA strain) exposed to untreated bottles/papers.
[2] Negative control: susceptible *An. quadriannulatus* exposed to treated bottles/papers.

An. funestus against a range of pyrethroid insecticides including deltamethrin.

This is the first time that *An. funestus* has been shown to exhibit pyrethroid insecticide resistance and our results highlight several important issues. Firstly, a molecular method for identification of members of the *An. funestus* group in southern Africa has proved its worth by demonstrating that four species of this group can be found resting inside houses or in exit traps. Clearly, *An. funestus* predominated and was implicated as a malaria vector in KZN. Questions also arise over the potential roles of *An. leesoni*, *An. parensis* and *An. rivulorum* collected from houses. Wilkes *et al*. (1996) reported *An. rivulorum* playing a minor role in malaria transmission in Tanzania. *Anopheles rivulorum* is the only member of the group that can sometimes be distinguished in the adult stage (without rearing F_1 progeny) using the morphological characteristic of an orange-brown thoracic coloration (Gillies & De Meillon, 1968). Because of the difficulty of identifying adult females of the *An. funestus* group, and because studies had shown most of the other species to be predominantly exophilic and zoophilic (Gillies & De Meillon, 1968), all previous work on indoor-resting *'An. funestus' sensu lato* has assumed that all such specimens are *An. funestus s.s.* Molecular methods are now available to assess other members of the *An. funestus* group for anthropophilic and endophilic behaviour and their role in malaria transmission.

The most important issue is the implication that these findings have for malaria control in southern Africa. Countries in this subregion have malaria control programmes based on vector control in combination with rapid case detection and effective treatment. Although all components of the programme are important for control of the disease, overall success is dependent on the reduction of transmission brought about by the control of the vector mosquitoes. This in turn is dependent on the availability of effective and safe insecticides that can be used in close association with the human population at risk. For many years, the insecticide of choice was DDT, and in some countries DDT is still preferred. In South Africa, since 1996, the malaria control programme had shifted to pyrethroids, regarded as more environmentally acceptable, not being bioaccumulative, not staining walls and, in some instances, having less excito-repellancy effects than DDT.

For anopheline mosquitoes, Malcolm (1988) concluded that physiological resistance to pyrethroids was not so widespread as had been suggested, and the risk of cross-resistance between

pyrethroids and DDT may have been over-emphasized. Thereafter, it was reported that a single mutation in the S6 transmembrane segment of domain II in the sodium channel sequence is associated with knockdown resistance (*kdr*) to both pyrethroids and DDT in the housefly, *Musca domestica* (Williamson *et al*., 1996) and the cockroach *Blattella germanica* (Miyazaki *et al*., 1996). This same mutation was detected in West African *An. gambiae* populations from Côte d'Ivoire that are resistant to pyrethroids (Martinez-Torres *et al*., 1998; Fanello *et al*., 1999) and a different substitution of the same amino acid results in pyrethroid resistance in East African *An. gambiae* (Ranson *et al*., 2000). *Anopheles gambiae* cross-resistance between DDT and several pyrethroids was demonstrated by Chandre *et al*. (1999b), apparently due to a *kdr* factor found in populations from six out of seven widely separated localities in West Africa. In our case, however, pyrethroid-resistant *An. funestus* remains fully susceptible to DDT, so a switch back to DDT would seem to be a solution to the present problem. If *kdr*-type cross-resistance is found in *An. funestus*, the choice of alternative insecticides for house spraying becomes very limited.

The PCR assay that has been developed to detect the sodium channel *kdr* protein mutation in *An. gambiae* (Martinez-Torres *et al*., 1998) has been tested on *An. funestus* individuals but only non-specific PCR products were obtained (Brooke *et al*., 1999). It is therefore necessary to sequence the same fragment of the sodium channel protein in *An. funestus* in order to assess whether any mutations are associated with the resistance.

At present, pyrethroid insecticides are the only option for impregnating bednets for malaria control and this situation must obviously be protected for as long as possible (Lengeler *et al*., 1996). A possible way forward is the careful consideration of the use of mixtures or rotational use of insecticides (Curtis, 1985; Curtis *et al*., 1993, 1998; Fanello *et al*., 1999) to slow down the widespread development of pyrethroid resistance or, worse, multiple resistance in vector populations. The malaria situation in most of tropical Africa is complicated by the presence of multiple species of vector mosquitoes that may have different responses to alternative insecticides. In South Africa, the presence of pyrethroid-resistant *An. funestus* and populations of *An. arabiensis* in which genes for resistance to carbamates and organophosphates have been detected (Hemingway, personal communication), underlines the importance of a multiple insecticide strategy. Therefore, careful monitoring of resistance in malaria vectors is clearly a priority in southern Africa, where most countries rely on vector control to reduce malaria transmission (WHO/SAMC, 1999).

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References

Brogdon, W.G. & McAllister, J.C. (1998) Simplification of adult mosquito bioassays through use of time-mortality determinations in glass bottles. *Journal of the American Mosquito Control Association*, **14**, 159–167.

Brooke, B.D., Hunt, R.H., Koekemoer, L.L., Dossou-Yovo, J. & Coetzee, M. (1999) Evaluation of a PCR assay for detection of pyrethroid insecticide resistance in the malaria vector species of the *Anopheles gambiae* complex (Diptera: Culicidae) suggests more than one mechanism of resistance. *Journal of the American Mosquito Control Association*, **15**, 565–568.

Brown, A.W.A. (1986) Insecticide resistance in mosquitoes: a pragmatic review. *Journal of the American Mosquito Control Association*, **2**, 123–140.

Brown, A.W.A. & Pal, R. (1973). *Insecticide Resistance in Arthropods*. 2nd edn. Monograph no. 38. World Health Organization, Geneva.

Chandre, F., Darriet, F., Manga, L., Akogbeto, M., Faye, O., Mouchet, J. & Guillet, P. (1999a) Status of pyrethroid resistance in *Anopheles gambiae* s.l. *Bulletin of the World Health Organization*, **77**, 230–234.

Chandre, F., Darriet, F., Manguin, S., Brengues, C., Carnevale, P. & Guillet, P. (1999b) Pyrethroid cross resistance spectrum among populations of *Anopheles gambiae s.s.* from Cote d'Ivoire. *Journal of the American Mosquito Control Association*, **15**, 53–59.

Chandre, F., Manguin, S., Brengues, C., Dossou Yovo, J., Darriet, F., Diabate, A., Carnevale, P. & Guillet, P. (1999c) Current distribution of a pyrethroid resistance gene (*kdr*) in *Anopheles gambiae* complex from West Africa and further evidence for reproductive isolation of the Mopti form. *Parassitologia*, **41**, 319–322.

Charlwood, J.D., Mendis, C., Thompson, R., Begtrup, K., Cuamba, N., Dgedge, M., Gamage-Mendis, A., Hunt, R.H., Sinden, R.E. & Hogh, B. (1998) *Cordon sanitaire* or *laissez faire*: differential dispersal of young and old females of the malaria vector *Anopheles funestus* Giles (Diptera: Culicidae) in southern Mozambique. *African Entomology*, **6**, 1–6.

Chavasse, D.C. & Yap, H.H., eds. (1997). *Chemical Methods for the Control of Vectors and Pests of Public Health Importance*. Document WHO/CTD/WHOPES/97.2. World Health Organization, Geneva.

Curtis, C.F. (1985) Theoretical models of the use of insecticide mixtures for the management of resistance. *Bulletin of Entomological Research*, **75**, 259–265.

Curtis, C.F. (ed) (1990) *Appropriate Technology in Vector Control*. CRC Press Incorporated, Florida.

Curtis, C.F., Hill, N. & Kasim, K. (1993) Are there effective resistance management strategies for vectors of human disease? *Biological Journal of the Linnean Society*, **48**, 3–18.

Curtis, C.F., Miller, J.E., Hassan, M.H., Kolaczinski, J.H. & Kasumba, I. (1998) Can anything be done to maintain the

effectiveness of pyrethroid impregnated bednets against malaria vectors? *Philosophical Transactions of the Royal Society, Series B,* **353**, 1769–1775.

De Meillon, B. (1936) The control of malaria in South Africa by measures directed against the adult mosquitoes in habitations. *Quarterly Bulletin of the Health Organization of the League of Nations*, **5**, 134–137.

De Meillon, B., Van Eeden, G.J., Coetzee, L., Coetzee, M., Meiswinkel, R., Du Toit, C.L.N. & Hansford, C.F. (1977) Observations on a species of the *Anopheles funestus* subgroup, a suspected exophilic vector of malaria parasites in northeastern Transvaal, South Africa. *Mosquito News*, **37**, 657–661.

Duchon, S., Finot, L., Chandre, F. & Guillet, P. (1997) Testing of pyrethroid impregnated papers on adult *Anopheles gambiae s.s.* mortality, KD rates and search for the diagnostic concentrations. ORSTOM/LIN technical report Nb.4 (unpublished report to WHO). Montpellier, November 1997.

Elissa, N., Mouchet, J., Riviere, F., Meunier, J.-Y. & Yao, K. (1993) Resistance of *Anopheles gambiae s.s.* to pyrethroids in Côte d'Ivoire. *Annales de la Societe Belge de Medecine Tropicale*, **73**, 291–294.

Fanello, C., Kolaczinski, J.H., Conway, D.J., Carnevale, P. & Curtis, C.F. (1999) The *kdr* gene in *Anopheles gambiae*: tests of non-pyrethroid insecticides and a new detection method for the gene. *Parassitologia*, **41**, 323–326.

Fontaine, R.E. (1983) The use of residual insecticides for the control of adult mosquitoes. *Integrated Mosquito Control Methodologies*, Vol. 1. *Experience and Components from Conventional Chemical Control* (ed. by M. Laird and J. W. Miles), pp. 49–81. Academic Press, London.

Freese, J.A., Markus, M.B. & Golenser, J. (1991) *In vitro* sensitivity of southern African reference isolates of *Plasmodium falciparum* to chloroquine and pyrimethamine. *Bulletin of the World Health Organization*, **69**, 707–712.

Gillies, M.T. & Coetzee, M. (1987) *Supplement to the Anophelinae of Africa South of the Sahara. Johannesburg: Publications of the South African Institute for Medical Research*, **55**, 1–143.

Gillies, M.T. & De Meillon, B. (1968) *The Anophelinae of Africa South of the Sahara. Johannesburg: Publications of the South African Institute for Medical Research*, **54**, 1–343.

Green, C.A. (1982) A cladistic analysis of mosquito chromosome data (*Anopheles* (*Cellia*) Myzomyia). *Journal of Heredity*, **73**, 2–11.

Green, C.A. & Hunt, R.H. (1980) Interpretation of variation in ovarian polytene chromosomes of *Anopheles funestus* Giles, *A. parensis* Gillies and *A. aruni*?. *Genetica*, **51**, 187–195.

Harrison, G. (1978). *Mosquitoes, Malaria and Man: a History of the Hostilities Since 1880*. John Murray, London.

Koekemoer, L.L., Lochouarn, L., Hunt, R.H. & Coetzee, M. (1999) Single-strand conformation polymorphism analysis for identification of four members of the *Anopheles funestus* (Diptera: Culicidae) group. *Journal of Medical Entomology*, **36**, 125–130.

Kouznetsov, R.L. (1977) Malaria control by application of indoor spraying of residual insecticides in tropical Africa and its impact on community health. *Tropical Doctor*, **7**, 81–91.

Laventure, S., Rabarison, P., Mouchet, J., Andrianaivolambo, L., Rakotaorivony, I., Rajaonarivelo, E. & Marrama, L. (1995) Paludisme: perspectives des recherches en entomologie medicale a Madagascar. *Cahiers Sante*, **5**, 406–410.

Lengeler, C., Cattani, J. & de Savigny, D. (eds) (1996). *Net Gain. A New Method for Preventing Malaria Deaths.* International Development Research Centre, Ottawa and World Health Organization, Geneva.

Lepers, J.P., Fontenille, D., Rason, M.D., *et al.* (1990) Transmission and epidemiology of newly transmitted falciparum malaria in the central highland plateaux of Madagascar. *Annals of Tropical Medicine and Parasitology*, **85**, 297–304.

Malcolm, C.A. (1988) Current status of pyrethroid resistance in anophelines. *Parasitology Today*, **4**, S13–S15.

Martinez-Torres, D., Chandre, F., Williamson, M.S., Darriet, F., Bergé, J.B., Devonshire, A.L., Guillet, P., Pasteur, N. & Pauron, D. (1998) Molecular characterization of pyrethroid knockdown resistance (*kdr*) in the major malaria vector, *Anopheles gambiae s.s. Insect Molecular Biology*, **7**, 179–184.

Mendis, C., Jacobsen, J.L., Gamage-Mendis, A., Bule, E., Dgedge, M., Thomson, R., Cuamba, N., Barreto, J., Begtrup, K., Sinden, R.E. & Hogh, B. (2000) *Anopheles arabiensis* and *An. funestus* are equally important vectors of malaria in Matola coastal suburb of Maputo, southern Mozambique. *Medical and Veterinary Entomology*, **14**, 171–180.

Miyazaki, M., Ohyama, K., Dunlap, D.Y. & Matsumura, F. (1996) Cloning and sequencing of the *para*-type sodium channel gene from susceptible and *kdr*-resistant German cockroaches (*Blattella germanica*) and house fly (*Musca domestica*). *Molecular and General Genetics*, **252**, 61–68.

Pampana, E.J. (1969) *A Textbook of Malaria Eradication*. 2nd edn. Oxford University Press, London.

Park Ross, G.A. (1936) Insecticide as a major measure in the control of malaria, being an account of the methods and organizations put into force in Natal and Zululand during the past six years. *Quarterly Bulletin of the Health Organization of the League of Nations*, **5**, 114–133.

Ranson, H., Jensen, B., Vulule, J.M., Wang, X., Hemingway, J. & Collins, F.H. (2000) Identification of a novel mutation in the voltage-gated sodium channel gene of *Anopheles gambiae* associated with resistance to pyrethroid insecticides. *Insect Molecular Biology*, in press.

Sharp, B.L. & Le Sueur, D. (1996) Malaria in South Africa – the past, the present and selected implications for the future. *South African Medical Journal*, **86**, 83–89.

Snounou, G., Viriyakosol, S., Zhu, X.P., Jarra, W., Pinheiro, L., Do Rosario, V.E., Thaithing, S. & Brown, K.N. (1993) High sensitivity of detection of human malaria parasites by the use of nested polymerase chain reaction. *Molecular and Biochemical Parasitology*, **61**, 315–320.

Touré, Y.T. (1982) Study of *Anopheles funestus* and *Anopheles gambiae s.l.* susceptibility to insecticides in a rural area of Sudan savanna in Mali. *Cahiers ORSTOM, Series Entomologie Medicale et Parasitologie*, **20**, 125–131.

Vulule, J.M., Beach, R.F., Atieli, F.K., Roberts, J.M., Mount, D.L. & Mwangi, R.W. (1994) Reduced susceptibility of *Anopheles gambiae* to permethrin associated with the use of permethrin-impregnated bednets and curtains in Kenya. *Medical and Veterinary Entomology*, **8**, 71–75.

Vulule, J.M., Beach, R.F., Atieli, F.K., Mount, D.L., Roberts, J.M. & Mwangi. R.W. (1996) Long-term use of permethrin-impregnated nets does not increase *Anopheles gambiae* permethrin tolerance. *Medical and Veterinary Entomology*, **10**, 71–79.

Vulule, J.M., Beach, R.F., Atieli, F.K., McAllister, J.C., Brogdon, W.G., Roberts, J.M., Mwangi, R.W. & Hawley, W.A. (1999) Elevated oxidase and esterase levels associated with permethrin tolerance in *Anopheles gambiae* from Kenyan villages using permethrin-impregnated nets. *Medical and Veterinary Entomology*, **13**, 239–244.

WHO (1975) *Manual on Practical Entomology in Malaria. Part II. Methods and Techniques.* World Health Organization, Geneva.

WHO (1998) Test procedures for insecticide resistance monitoring in

malaria vectors, bio-efficacy and persistence of insecticides on treated surfaces. Document WHO/CDS/CPC/MAL/98.12. World Health Organization, Geneva.

WHO/SAMC (1999) Intensive vector control activities begin throughout southern Africa. *Southern Africa Malaria Control, Information for Action Leaflet. Malaria Vector Update*, **1** (1), 1–2. World Health Organization, Harare.

Wilkes, T., Matola, Y.G. & Charlwood, J.D. (1996) *Anopheles rivulorum*, a vector of human malaria in Africa. *Medical and Veterinary Entomology*, **10**, 108–110.

Williamson, M.S., Martinez-Torres, D., Hick, C.A. & Devonshire, A.L. (1996) Identification of mutations in the housefly *para*-type sodium channel gene associated with knockdown resistance (*kdr*) to pyrethroid insecticides. *Molecular and General Genetics*, **252**, 51–60.

Wirtz, R.A., Zavala, F., Charoenvit, Y., Cambell, G.H., Burkot, T.R., Schneider, I., Esser, K.M., Beaudoin, R.L. & Andre, G.R. (1987) Comparative testing of *Plasmodium falciparum* sporozoite monoclonal antibodies for ELISA development. *Bulletin of the World Health Organization*, **65**, 39–45.

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